O005	Vikram Reddy	Smart Watch	2	15000.00
O006	Anjali Gupta	Laptop	1	52000.00
O007	Suresh Nair	Keyboard	4	1500.00
O008	Kavita Joshi	Mouse	6	800.00

III. Generate and execute the following queries

1. Change the size of customer_name to 55. (2marks)
2. Increase the quantity of products to 3 which are less than 3. (2marks)
3. Delete the product whose price is less than 1000. (2marks)
4. Find the total number of laptops ordered. (2marks)
5. Find the total quantity of each ordered product. (2marks)

Q6. Sports Club System

I. Create a table named MEMBER with the following structure. (5marks)

Column Name	Data Type	Constraints
member_id	VARCHAR(10)	PRIMARY KEY
member_name	VARCHAR(50)	NOT NULL
sport	VARCHAR(30)	NOT NULL
membership_type	VARCHAR(20)	
fees	DECIMAL(8,2)	
age	INT	
joining_date	DATE	

II. Insert the following data into the MEMBER table. (5marks)

member_id	member_name	sport	membership_type	fees	age	joining_date
MB001	Rohit Sharma	Cricket	Annual	15000.00	28	2023-01-15
MB002	Saina Nehwal	Badminton	Monthly	2500.00	25	2023-06-20
MB003	Virat Kohli	Cricket	Annual	15000.00	30	2022-08-10
MB004	Mary Kom	Boxing	Quarterly	8000.00	35	2023-03-25
MB005	Neeraj Chopra	Athletics	Annual	12000.00	26	2023-02-18
MB006	PV Sindhu	Badminton	Annual	14000.00	27	2022-11-05
MB007	Sunil Chhetri	Football	Monthly	3000.00	32	2023-07-12
MB008	Hima Das	Athletics	Quarterly	7500.00	24	2023-05-08

III. Generate and execute the following queries

1. Modify the size of sport to 35. (2marks)
2. Update the membership type of member 'Saina Nehwal' to 'Annual' and fees to 14000.00. (2marks)
3. Delete all members from the MEMBER table who have 'Monthly' membership type. (2marks)
4. Display the details of members with ‘Annual’ membership in the alphabetical order of their names. (2marks)
5. Find the total membership of each type. (2marks)

Q7. Hospital Management System

I. Create a table named PATIENT with the following structure. (5marks)

Column Name	Data Type	Constraints
patient_id	VARCHAR(10)	PRIMARY KEY
patient_name	VARCHAR(50)	NOT NULL
age	INT	
gender	VARCHAR(10)	
disease	VARCHAR(50)	
doctor_name	VARCHAR(50)	

II. Insert the following data into the PATIENT table. (5marks)

patient_id	patient_name	age	gender	disease	doctor_name
P101	Rajesh Mehta	45	Male	Diabetes	Dr. Sharma
P102	Anita Desai	32	Female	Fever	Dr. Patel
P103	Sunil Kapoor	58	Male	Heart Disease	Dr. Kumar
P104	Priya Nair	28	Female	Dengue	Dr. Sharma
P105	Mohan Lal	62	Male	Hypertension	Dr. Verma
P106	Seema Roy	40	Female	Arthritis	Dr. Kumar
P107	Arjun Singh	35	Male	Fever	Dr. Patel
P108	Kavita Jain	50	Female	Diabetes	Dr. Sharma

III. Generate and execute the following queries

1. Add a new column admission_date of date data type. (2marks)

2. Update the doctor name for patient 'Anita Desai' to 'Dr. Verma'. (2marks)
3. Display the details of patients above 50 years. (2marks)
4. Find the average age of patients in the table. (2marks)
5. Display the number of male and female patients. (2marks)

Q8. Movie Theater System

I. Create a table named MOVIE with the following structure. (5marks)

Column Name	Data Type	Constraints
movie_id	VARCHAR(10)	PRIMARY KEY
title	VARCHAR(80)	NOT NULL
genre	VARCHAR(30)	
director	VARCHAR(50)	
duration	INT	
rating	DECIMAL(2,1)	
release_date	DATE	

II. Insert the following data into the MOVIE table. (5marks)

movie_id	title	genre	director	duration	rating	release_date
M001	Inception	Sci-Fi	Christopher Nolan	148	8.8	2010-07-16
M002	The Shawshank Redemption	Drama	Frank Darabont	142	9.3	1994-09-23
M003	The Dark Knight	Action	Christopher Nolan	152	9.0	2008-07-18

M004	Forrest Gump	Drama	Robert Zemeckis	142	8.8	1994-07-06
M005	Interstellar	Sci-Fi	Christopher Nolan	169	8.6	2014-11-07
M006	The Godfather	Crime	Francis Coppola	175	9.2	1972-03-24
M007	Avatar	Sci-Fi	James Cameron	162	7.9	2009-12-18
M008	Titanic	Romance	James Cameron	194	7.9	1997-12-19

III. Generate and execute the following queries

1. Add a default value constraint to the column rating with value 4.0. (2marks)
2. Update the ratings which are less than 8 to 8.0. (2marks)
3. Display the details of movies in Sci-Fi genre. (2marks)
4. What is the average rating of movies in the table. (2marks)
5. Sort the details of movies in the highest order of rating. (2marks)

Q9. Course Registration System

I. Create a table named REGISTRATION with the following structure. (5marks).

Column Name	Data Type	Constraints
reg_id	VARCHAR(10)	PRIMARY KEY
student_name	VARCHAR(50)	NOT NULL
course_name	VARCHAR(60)	NOT NULL
course_code	VARCHAR(10)	
credits	INT	

semester	INT	
reg_date	DATE	

II. Insert the following data into the REGISTRATION table. (5marks)

reg_id	student_ name	course_name	course_code	credits	semester	reg_date
R001	Amit Sharma	Database Systems	CS301	4	3	2024-08-01
R002	Neha Gupta	Operating Systems	CS302	4	3	2024-08-02
R003	Rahul Verma	Data Structures	CS201	3	2	2024-08-01
R004	Priya Singh	Web Development	CS303	3	3	2024-08-03
R005	Vikram Reddy	Database Systems	CS301	4	3	2024-08-02
R006	Anjali Patel	Computer Networks	CS304	4	3	2024-08-04
R007	Suresh Kumar	Data Structures	CS201	3	2	2024-08-01
R008	Kavita Iyer	Machine Learning	CS401	4	4	2024-08-05

III. Generate and execute the following queries

1. Add a default value constraint to the column credits with value 3. (2marks)
2. Update the credits for course 'Web Development' to 4. (2marks)
3. Display the details of courses in semester 3 and credit 4. (2marks)
4. How many courses are there with credit 4. (2marks)

5. Find the total number of credits registered by student 'Amit Sharma'. (2marks)

Q10. Vehicle Showroom System

I. Create a table named VEHICLE with the following structure. (5marks).

Column Name	Data Type	Constraints
vehicle_id	VARCHAR(10)	PRIMARY KEY
model_name	VARCHAR(50)	NOT NULL
brand	VARCHAR(30)	NOT NULL
color	VARCHAR(20)	
price	DECIMAL(10,2)	
mfg_year	INT	
fuel_type	VARCHAR(15)	

II. Insert the following data into the VEHICLE table. (5marks)

vehicle_id	model_name	brand	color	price	mfg_year	fuel_type
V001	City	Honda	White	1250000.00	2023	Petrol
V002	Creta	Hyundai	Red	1580000.00	2024	Diesel
V003	Swift	Maruti	Blue	850000.00	2023	Petrol
V004	Seltos	Kia	Black	1680000.00	2024	Diesel
V005	Venue	Hyundai	Silver	1180000.00	2023	Petrol
V006	Baleno	Maruti	Grey	920000.00	2024	Petrol
V007	XUV700	Mahindra	White	2150000.00	2024	Diesel